

Infosys_HotelRevAI – AI Driven Revenue Analysis for Hotels

Milestone 1: Orientation and Data Loading

Progress Report – Milestone 1

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1. Introduction

The Hotel Revenue AI Analysis Project is a smart system that analyzes hotel booking and revenue data using Artificial Intelligence and data analytics. The system helps hotel management understand revenue trends, booking patterns, and customer behavior. It also predicts future revenue and provides suggestions to improve revenue on the basis of performance.

2. Problem Statement

Hotels generate large volumes of booking and revenue data. Manual analysis of this data is difficult and may lead to incorrect decisions. Hotels often face challenges such as:

- Identifying revenue trends
- Predicting future revenue
- Detecting revenue decline
- Understanding customer booking patterns

An AI-based system is required to automate revenue analysis and provide improvement suggestions.

3. Objectives

- Analyze hotel booking and revenue data
- Calculate profit and revenue trends
- Identify seasonal booking patterns
- Predict future revenue using AI
- Provide revenue improvement suggestions
- Generate visual reports and dashboards

4. Scope of the Project

The project focuses on analyzing historical hotel booking data and generating revenue insights. It includes data processing, revenue analysis, prediction models, visualization dashboards, and suggestion generation. The system can be extended for real-time booking analysis in the future.

5. System Modules

- **Data Input Module**
Collects hotel booking and revenue data.
- **Data Processing Module**
Cleans data, removes errors, and prepares data for analysis.

- **Revenue Analysis Module**
Calculates revenue, profit, and booking trends.
- **AI Prediction Module**
Uses machine learning models to calculate the revenue.
- **Revenue Suggestion Module**
Compares current revenue with past data and provides improvement suggestions if revenue decreases.
- **Visualization and Report Generation Module**
After all the operation it will generate the report which includes the revenue,etc. and visual representation like Dashboard,Graphs,Charts. If it has a revenue decrement then give suggestions also.

6. Technology Used

- Programming: Python / Java
- Frontend: HTML, CSS, JavaScript
- Backend: Flask / Spring Boot
- Database: MySQL / CSV
- AI Libraries: Pandas, NumPy, Matplotlib
- Data Visualization & Dashboard Creation: Power BI / Tableau / Python Visualization Tools

7. Methodology

- 1. Data Collection**
Collect hotel booking and revenue dataset from database, CSV, or Excel files.
- 2. Data Preprocessing**
Clean the dataset by removing errors, handling missing values, and formatting data properly.
- 3. Data Analysis**
Analyze booking patterns, seasonal trends, customer behavior, revenue, and profit details.
- 4. AI Revenue Prediction**
Apply machine learning models to predict future revenue and identify performance trends.
- 5. Revenue Comparison**
Compare current revenue with past revenue performance to detect increase or decrease.
- 6. Suggestion Generation**
If revenue decreases, the system generates improvement suggestions such as pricing adjustments, promotional offers,Improving customer service or marketing strategies.
- 7. Report and Dashboard Generation**
The system generates a final report and interactive dashboard showing:
 - Revenue details
 - Profit analysis
 - Booking trends
 - Prediction results
 - Improvement suggestions (if revenue decline is detected)

8. Features

- Automated revenue and profit calculation
- AI-based revenue prediction
- Booking trend analysis
- Revenue improvement suggestions
- A Report Generated with Dashboard visualization

9. Advantages

- Saves time and reduces manual effort
- Provides accurate revenue insights
- Helps improve business strategies
- Supports data-driven decision making

10. Conclusion

The Hotel Revenue AI Analysis Project is an efficient system that analyzes hotel revenue data using AI techniques. It calculate revenue and predict revenue trends and provides improvement suggestions, helping hotel management make better business decisions and improve profitability.